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A review on the ductility design method of fiber-reinforced polymer bar and future prospects

Yunbo Xu¹ | Yu Zhang¹ | Haitang Zhu¹ | Danying Gao² | Daotian Qin¹ | Liangping Zhao¹

¹College of Civil Engineering, Henan University of Engineering, Zhengzhou, Henan, China

²School of Civil Engineering, Zhengzhou University, Zhengzhou, Henan, China

Correspondence

Yu Zhang, College of Civil Engineering, Henan University of Engineering, Zhengzhou, 450001, Henan, China.
Email: zy201811211030163@163.com

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Abstract

In this paper, the ductility design methods of fiber-reinforced polymer (FRP) bars were reviewed. It was observed that the graded fracture theory was typically used as the ductility design method of hybrid fiber-reinforced polymer (HFRP) bar. However, the ductile HFRP bar designed based on the graded fracture theory had the inherent defects of low modulus of elasticity, high yield strain, and post-yielded sudden drop in stress, which prevented its large-scale application in civil engineering. In order to eliminate these deficiencies, the authors proposed a novel design concept for a single-type FRP bar. This novel single-type FRP bar consisted of highly aligned discontinuous fiber and continuous fiber. The failure mode of this discontinuous/continuous single-type FRP bar was different from that of the ductile HFRP bar designed based on the graded fracture theory of composite. The tensile ductility of discontinuous/continuous single-type FRP bar originated from the debonding and stable pull-out of the discontinuous fiber layer under increasing load. As a result, the post-yielded sudden drop in stress can be removed for the ductile HFRP bar designed based on the graded fracture theory of composite. In addition, the yield strain can be controlled by adjusting the length of discontinuous fiber layer. In addition, the design configuration, innovative production process, and corresponding theoretical calculations of this novel single-type FRP bar will be presented in the future.

Highlights

- The ductility design methods of fiber-reinforced polymer bars were reviewed.
- Deficiencies of ductile HFRP bar composed of continuous fibers were reported.
- A novel discontinuous/continuous single-type FRP bar was foreseen.

KEYWORDS

continuous fiber, discontinuous fiber, ductility design method, fiber-reinforced polymer bar, graded fracture theory

1 | INTRODUCTION

The corrosion of steel bars embedded in concrete has brought enormous losses to structural buildings, as shown in Figure 1, and this has become a consensus among all humanity.^{1–4} Although some remedies have been suggested by researchers to alleviate the current situation,^{5–9} as shown in Figure 2, the corrosion of steel bars cannot be completely eliminated in situ environment.^{10,11} Hence, the fiber-reinforced polymer (FRP) bars have developed into a potential alternative, which attracts the extensive attention due to their essential advantages of corrosion resistance, no magnetization, lightweight, high strength, and so on.^{12–15} FRP bar is manufactured by pultruding or braiding multiple continuous fiber bundles impregnated with polymer resin and then thermally treated. The commercial continuous fibers include aramid fiber, basalt fiber, carbon fiber, glass fiber, natural fiber, and so on. The polymer resin can be divided into thermosetting and thermoplastic

resin. The thermosetting resin is typically used as the matrix phase of FRP bars, containing epoxy resin, vinyl resin, and polyester resin.¹¹

FRP bar-reinforced concrete was first realized in the mid-20th century. In 1986, the world's first FRP bar-reinforced highway bridge was built in Germany. Subsequently, this trend was spread throughout Japan, Europe, and North America.¹¹ However, until now, the characteristics of brittle failure and low tensile modulus of elasticity for single-type FRP bars still hinder their large-scale promotion and application in civil engineering field.¹⁶ The stress-strain responses of common single-type FRP bars under tensile load are shown in Figure 3. In addition, the typical mechanical and physical properties of common single-type FRP bars are presented in Table 1.

In recent decades, abundant researchers have devoted themselves to developing high-performance FRP bars, wherein the hot topics mainly focus on solving the tensile ductility of single-type FRP bars and improving their modulus of elasticity. The method to achieve

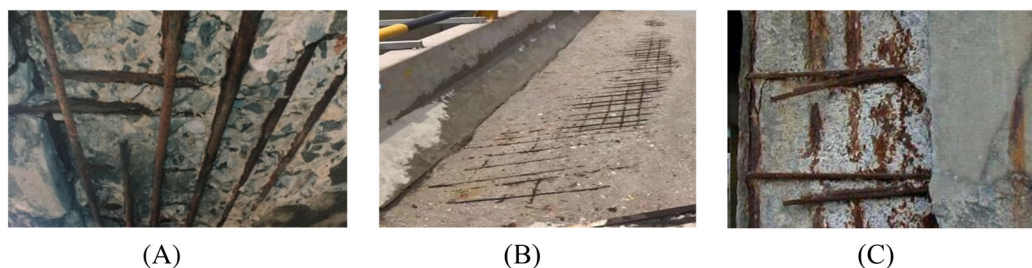


FIGURE 1 Detachment of concrete protective layer resulting from corrosion of steel bar. (A) Galvanized steel bar. (B) Wrapping epoxy resin. (C) Wrapping composite layer.

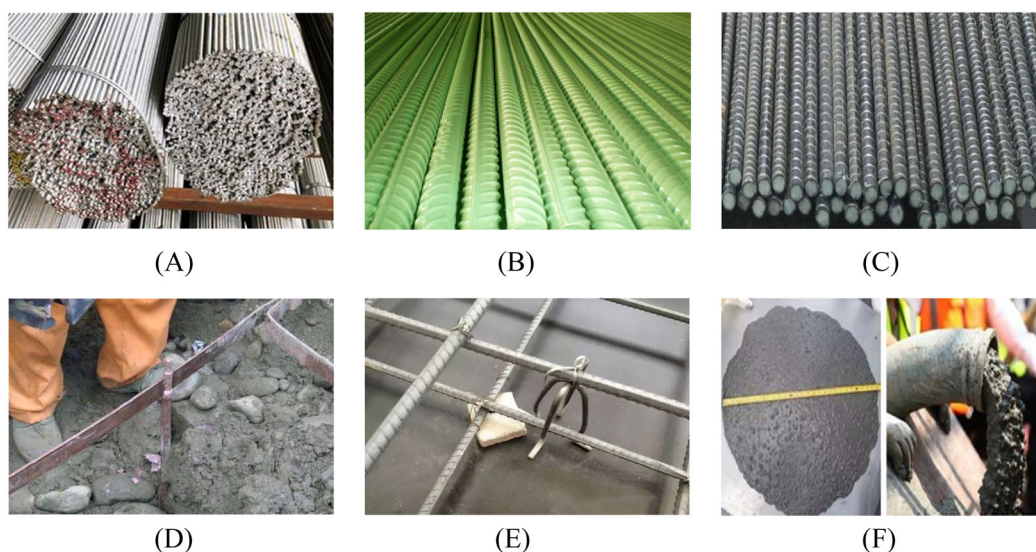


FIGURE 2 Corrosion prevention measures for steel bars in structural concrete. (A) Galvanized steel bar. (B) Wrapping epoxy resin. (C) Wrapping composite layer. (D) Cathodic protection. (E) Increasing depth of cover layer. (F) High durability concrete.

high-performance FRP bar is to use hybrid fibers as the reinforced phase of polymer resin. Controlling the volume ratio of hybrid fibers realizes their successive fracture under tensile load, which can remove the brittle failure of single-type FRP bar and enhance its low modulus of elasticity.^{17–19} The high-performance FRP bar is also known as hybrid fiber-reinforced polymer (HFRP) bar. The reinforced phase of HFRP bar is composed of two or more types of continuous fibers with different elongations. The high modulus fiber (low elongation fiber) that makes up hybrid fibers first reaches its ultimate strain and breaks out under tensile load. The released load from failed fiber is transferred to the remaining low-modulus fiber (high elongation fiber) through the matrix resin among fibers. Subsequently, the stress state of HFRP bar can be divided into two situations. When the volume ratio of hybrid fibers is unsuitable, the failure of high-modulus fiber is closely followed by the destruction of HFRP bar. When the volume ratio of hybrid fibers is controlled, the HFRP bar shows strain hardening to undertake the further increasing load until

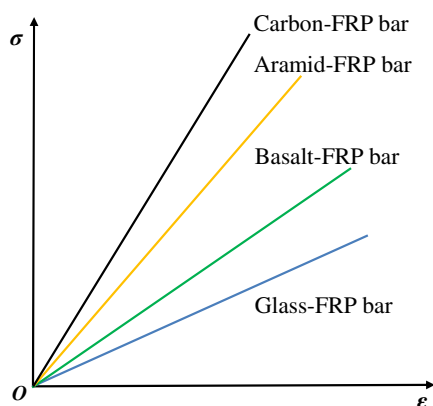


FIGURE 3 Typical tensile stress–strain curves of normal single-type FRP bars.

the ultimate state of HFRP bar is achieved.^{20–22} The corresponding tensile stress–strain responses can be found in the following text. Therefore, it can be seen that the working mechanism of HFRP bar is to use the difference in tensile moduli (elongation) of hybrid fibers under tensile load to realize their graded fracture and obtain the ductility of HFRP bar.

In particularly, the reports on the tensile ductility of hybrid fibers-reinforced polymer laminates grew vigorously around the 1970s, whereas applying this method to FRP bar was decades later.^{23–27} In this review, although HFRP laminates have a longer development history, HFRP bar is the critical object. Based on the type of fiber, continuous, and discontinuous states of fiber in HFRP bar, the ductility design of HFRP bar can currently be divided into four categories, including: (1) mingling of two or more continuous fibers, (2) mingling of continuous fiber, and steel bar/wire, (3) mingling of continuous fiber and the chopped-fiber randomly dispersed into polymer resin, (4) mingling of highly aligned discontinuous fiber and continuous fiber. The aforementioned design methods of ductile HFRP bar and the research results will be reviewed in the following paragraphs, and a comprehensive evaluation will be made at the end of this part.

1.1 | Mingling of two or more continuous fibers

Mingling of two or more continuous fibers to reinforce polymer resin is the most commonly used method to solve the brittle failure of single-type fiber-reinforced polymer bar under tensile load. According to the production processes of FRP bar, HFRP bar with two or more continuous fibers can typically be separated into the pultrusion-winding HFRP bar and the pultrusion-braiding HFRP bar. The combination of pultrusion and winding

TABLE 1 Mechanical and physical properties of single-type FRP bar under tensile load.

Bar type	Steel bar	Steel strand	Glass-FRP bar	Carbon-FRP bar	Aramid-FRP bar	Basalt-FRP bar
E (GPa)	210	180 ~ 200	35 ~ 65	120 ~ 580	40 ~ 125	40 ~ 70
σ_y (MPa)	280 ~ 420	1050 ~ 1400	–	–	–	–
σ_u (MPa)	490 ~ 700	1400–1890	480 ~ 1600	600 ~ 3700	1200 ~ 2550	800 ~ 1800
ϵ_u (%)	>10.0	>4.0	1.2 ~ 3.1	0.5 ~ 1.7	1.9 ~ 4.4	2.0 ~ 2.5
ρ (kg/m ³)	7850	7850	1250 ~ 2100	1500 ~ 1600	1250 ~ 1400	1900 ~ 2100
α_L (10 ^{–6} /°C)	11.7	11.7	3.3 ~ 5.6	–4.0 ~ 0.0	–3.3 ~ –1.1	9.0 ~ 12.0
α_T (10 ^{–6} /°C)	11.7	11.7	11.7 ~ 12.8	41.0 ~ 58.0	33.3 ~ 44.4	21.0 ~ 22.0
R_s (%)	–	3.0	1.8	1.0 ~ 3.0	7.0 ~ 20.0	10.0 ~ 16.0

Note: E denotes modulus of elasticity. σ_y , σ_u , and ϵ_u signify the yield strength, ultimate strength, and ultimate strain, respectively. ρ stands for density of bar. α_L and α_T represent the longitudinal and transverse temperature expansion coefficients of bar, respectively. R_s is representative for stress relaxation rate.

processes can produce the dented spiral ribs on the surface of HFRP bar so that it has a good bond behavior with concrete, as shown in Figure 4A. Meanwhile, the hybrid fibers in the pultrusion-winding HFRP bar are as straight as possible, which fully maintains the mechanical properties of the fibers themselves. As shown in Figure 4B, the combination of pultrusion and braiding processes can form a woven structure on the outer of pultruded FRP core, which will realize a great roughness on the surface and a higher tensile elongation for pultrusion-braiding HFRP bar.^{28–30} At the initial stage of load application, the pultrusion-braiding HFRP bar shares a better bond behavior together with concrete under pull-out load, but the pultruded core of HFRP bar embedded in concrete first fails with the increase in loading. Subsequently, the misaligned fibers inside the braided shell are gradually straightened under tensile load, accompanied by radial shrinkage of HFRP bar during this ongoing process, which will result in a sharp degradation of its bond behavior with concrete. Therefore, the intrinsic advantages of simple operation, high flexibility, strong reproducibility, high material utilization, high production efficiency, and superior product quality make the pultrusion-winding processes of HFRP bar become the most potential technology comparing to the pultrusion-braiding processes.

In addition, based on the distribution pattern of hybrid fibers inside HFRP bars, they can also be divided into HFRP bar with centralized arrangement of high modulus fiber (low elongation fiber), and HFRP bars with interlayer, intralayer, or intrayarn distribution between high and low moduli fibers (low and high elongation fibers), as shown in Figure 5A–D. Research results indicated that the greater the dispersion of high modulus fiber (low elongation fiber) inside HFRP bar, the better the tensile ductility of HFRP bar.³¹ Until now, there is little research on the HFRP bar with a dispersion distribution of high modulus fiber (low elongation fiber) inside the intrayarn of low modulus fiber (high elongation fiber) due to limited production technology of fiber bundle.

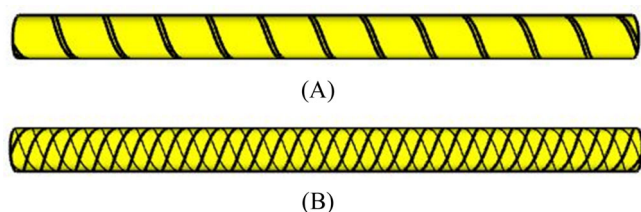


FIGURE 4 Surface configurations of HFRP bars prepared by different processes. (A) Pultrusion-winding processes. (B) Pultrusion-braiding processes.

1.1.1 | HFRP bars prepared by pultrusion-winding processes

The effects of volume ratio and distribution pattern of hybrid fibers on the tensile ductility of HFRP bar have been reported by many researchers, wherein the reports on carbon/glass-HFRP bar accounted for the main part. Yu et al.³¹ pointed out that the tensile ductility of carbon/glass-HFRP bar could be realized based on the graded fracture theory of composites; the lower carbon/glass ratio in the carbon/glass-HFRP bar, the more significant the tensile ductility of carbon/glass-HFRP bar. Zhao et al.³² also concluded that when the carbon/glass ratio in carbon/glass-HFRP bar was lower than the critical value determined by the graded fracture theory, the carbon/glass-HFRP bar presented second-order failure or plastic deformation under tensile load, otherwise it showed brittle failure. Similarly, You et al.^{33–35} claimed that the carbon/glass-HFRP bar would exhibit brittle failure if the carbon/glass ratio was beyond a critical carbon/glass ratio. Xu et al.^{21,36} tested a carbon/glass-HFRP bar with an approximate 6:1 carbon/glass ratio. The experiment results indicated that the mechanical properties of carbon/glass-HFRP bar were nearly approached to those of carbon-FRP bar, and the tensile ductility of carbon/glass-HFRP bar was not observed due to a higher carbon/glass ratio. Cao et al.³⁷ studied on the tensile properties of carbon/glass-HFRP bars with different carbon/glass ratios. The conclusions confirmed that the tensile stress-strain response of carbon/glass-HFRP bars presented a stepped development trend, which reflected the tensile ductility of carbon/glass-HFRP bar to some extent. Xu et al.³⁸ highlighted that HFRP bar could obtain a similar yield strain and plastic deformation to those of steel bar by mingling fibers with different moduli. The conclusion was consistent with the view point of Belarbi et al.³⁹

He et al.⁴⁰ studied on the hybrid effect of carbon/glass-HFRP bar by intra-layer mingling of carbon fiber and glass fiber, and the yield phenomenon of carbon/glass-HFRP bar with a 1:6 carbon/glass ratio was obviously observed under tensile load. Yang et al.⁴¹ claimed that the tensile stress-strain response of carbon/glass-HFRP bar with an intra-layer mingling distribution of carbon fiber and glass fiber presented a yield platform similar to that of steel bar. Kumar et al.⁴² also performed a similar tensile test of carbon/glass-HFRP bar. Duan et al.⁴³ prepared carbon/glass-HFRP bars with the interlayer or intra-layer mingling of carbon fiber and glass fiber to study their tensile fatigue behaviors in the application of civil engineering. Garbacz et al.⁴⁴ simulated the tensile properties of carbon/glass-HFRP bar and carbon/basalt-HFRP bar based on the distribution pattern of hybrid fibers. The simulation results revealed that when

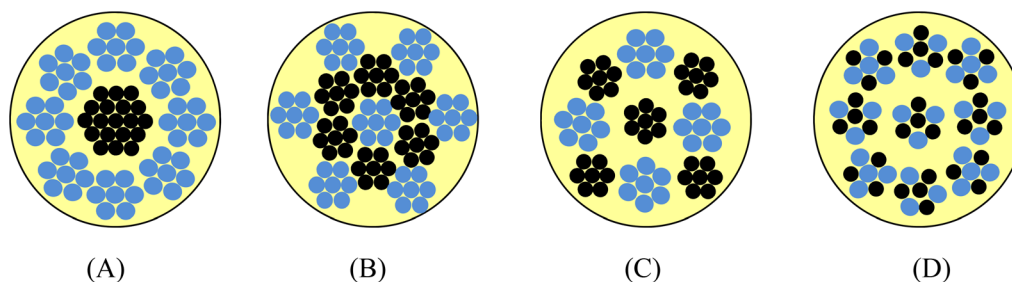


FIGURE 5 Distribution patterns of hybrid fibers inside HFRP bars. (Black, blue, and yellow represent high-modulus fiber, low-modulus fiber, and polymer resin, respectively.) (A) Centralization. (B) Interlayer. (C) Intra-layer. (D) Intra-yarn.

carbon fiber was mixed at a lower volume fraction, it was better to locate in the near-surfaces of carbon/glass-HFRP bar and carbon/basalt-HFRP bar, otherwise located in their core area. Deng et al.⁴⁵ designed three carbon/glass-HFRP bars with different arrangement patterns of carbon fiber, and the test results showed that their failure modes were splitting, belonging to brittle failure.

Limited by the tensile modulus (or elongation) of normal glass fiber (E-glass fiber), the high-strength glass fiber (S-glass fiber) was selected to mingle with carbon fiber or aramid fiber to obtain the high ductile HFRP bars by some researchers. The common types of glass fiber and their mechanical properties are presented in Table 2. Two types of carbon/S-glass HFRP bars with centralization and intra-layer arrangements of carbon fiber were separately tested by Liang et al.^{46,47} Qiu et al.⁴⁸ fabricated a micro-composite hybrid rod using Kevlar fiber as high-modulus fiber and S-glass fiber as low-modulus fiber to understand its tensile properties. The results confirmed that a bilinear tensile stress-strain response was presented, whereas a similar to “knee point” with a sharp drop in stress appeared, as shown in the figure in the following section.

As a result of non-alkali resistance and low modulus of elasticity for normal glass fiber (E-glass fiber), the aramid fiber with a higher modulus of elasticity or the environmentally friendly basalt fiber was chosen to mingle with carbon fiber. Ali et al.^{49,50} tested the tensile properties of carbon/basalt- and carbon/glass-HFRP bars based on different volume ratios and distribution patterns of carbon fiber. It was observed that the tensile stress-strain response of linear elasticity for single-type FRP bar could be modified by hybrid fibers with different moduli, obtaining the ductility features for carbon/basalt- and carbon/glass-HFRP bars under tensile load. Protchenko et al.⁵¹ replaced a part of basalt fiber with carbon fiber to prepare carbon/basalt-HFRP bar, aiming to enhance the mechanical properties of basalt-FRP bar and the cost performance of carbon-FRP bar. Elsayed et al.⁵² applied

TABLE 2 Tensile properties of common glass fibers.

Fiber type	E (GPa)	σ_u (MPa)	ϵ_u (%)
E-glass fiber	65	1600	3.1
S-glass fiber	88	5000	5.5
S3-glass fiber	99	3300	—

Note: E is modulus of elasticity for glass fiber. σ_u and ϵ_u denote the ultimate strength and strain of glass fiber, respectively.

carbon/aramid- and carbon/glass-HFRP bars to substitute for carbon-FRP bar or glass-FRP bar to reinforce concrete members, indicating that the carbon/aramid- or carbon/glass-HFRP bars-reinforced concrete members exhibited a definite flexural ductility. Wang et al.⁵³ evaluated the tensile ductility of carbon/basalt-HFRP bars with different carbon/basalt ratios. It was reported that the tensile stress-strain response of carbon/basalt-HFRP bar did not exhibit significant multilinearity. The reason could be attributed to a higher carbon/basalt ratio, resulting in the premature failure of carbon/basalt-HFRP bar after the carbon fiber failed. However, the correctness cannot be verified due to missing the mechanical properties of constituent material in the report. Besides, Belarbi et al.⁵⁴ provided a smart FRP bar, consisting of a pultruded core of carbon fiber and a braided shell of carbon fiber, to realize its tensile ductility and health-monitoring capability. The tensile ductility of this smart FRP bar came from the graded fracture of different fiber layers.

Simultaneously, some researchers also explored the effect of the mingling of no less than three types of fibers on the tensile ductility of HFRP bar. Krishnamoorthy et al.⁵⁵ manufactured a new type of HFRP bar, composed of a hybrid core of three types of carbon fibers and an outer layer of helical fiber, to ensure its initial modulus and yield at a lower tensile strain. The working mechanism was that the helical fiber layer could undergo the load released from the failed hybrid core under a gradual increasing load. This procedure could ensure that the novel HFRP bar underwent a large deformation before failure. Bakis et al.⁵⁶ demonstrated the tensile ductility of

seven pultruded HFRP bars by mingling three types of fibers each other, including: carbon/glass-HFRP bars with different carbon/glass ratios and distribution patterns of carbon fiber, carbon/glass/PVA-HFRP bar, and carbon/aramid/PVA-HFRP bars with different types of polymer resin. Additionally, Behnam et al.^{57,58} prepared three types of HFRP bars, including: (a) interlayer mingling of carbon fiber (intermediate modulus, IM) and aramid fiber, (b) interlayer mingling of carbon fiber (IM), carbon fiber (low modulus, LM) and aramid fiber, and (c) interlayer mingling of carbon fiber (IM), carbon fiber (LM), aramid fiber, and glass fiber. Their study identified that the tensile stress-strain responses presented multiple yield points and a large plastic deformation for HFRP bars by mingling no less than three types of continuous fiber.

In order to mitigate the problem of weak interface brought by mingling different diameters of monofilament fibers, Szmigiera et al.⁵⁹ tested the tensile properties of carbon/basalt-HFRP bar incorporated with silica nanoparticles. Due to the high economic cost of inorganic fiber, Safaan et al.⁶⁰ mingled glass fiber with nylon rope to prepare glass/nylon-HFRP bar, and the volume fraction of nylon rope ranged from 0% to 20%. The conclusions indicated that the tensile stress-strain responses of all glass/nylon-HFRP bars were completely linear elastic, resulting from that the volume fraction of nylon rope was too low to undergo the load released by the failed glass fiber.

1.1.2 | HFRP bar prepared by pultrusion-braiding processes

The advantages of surface roughness and high straightening deformation for a braided sleeve of HFRP bar, contributing to bond with concrete, and engendering a high tensile ductility, have attracted numerous attentions, as shown in Figure 4B. The braided sleeve of HFRP bar was mainly made of aramid fiber due to its superior mechanical properties relative to those of glass fiber. Harris et al.⁶¹ used pultrusion and braiding processes to prepare a novel carbon/aramid-HFRP bar, consisting of a pultruded core and a braided sleeve, which made the carbon/aramid-HFRP bar possess a bilinear tensile stress-strain response and an initial modulus of elasticity similar to that of steel bar. Hampton et al.^{62–66} pointed out that the tensile stress-strain response of second-generation HFRP bar, consisting of a pultruded core of high modulus carbon fiber and a braided sleeve made of aramid fiber, displayed a bilinear or trilinear development trend, which indicated the plastic failure mode of second-generation carbon/aramid-HFRP bar under

tensile load. The pultruded core of carbon fiber provided an initial modulus for carbon/aramid-HFRP bar, while the strain hardening of the braided sleeve made of aramid fiber realized the tensile ductility of carbon/aramid-HFRP bar. Somboonsong et al.⁶⁷ performed a similar experiment to the aforementioned content.^{62–66} Rosa et al.⁶⁸ prepared a carbon/aramid-HFRP bar, composed of a pultruded core mingled by four types of carbon fibers and a braided shell made of aramid fiber. The plastic deformation of carbon/aramid-HFRP bar was reflected by the obtaining multi-stage stress-strain response under tensile load. Additionally, in order to evaluate the effect of hybrid braided sleeve on plastic deformation of HFRP bar, Park et al.⁶⁹ and Won et al.^{70–72} carried out tensile tests on carbon/aramid-HFRP bar, carbon/glass-HFRP bar, and carbon/aramid/glass-HFRP bar, consisting of a pultruded core of carbon fiber and a braided sleeve of aramid fiber, a pultruded core of carbon fiber and a braided sleeve of glass fiber, and a pultruded core of carbon fiber and a braided sleeve of aramid fiber mingled with glass fiber, respectively. It was found that a yield strain approaching 1.2% and a strain hardening stage were observed in the experimental tensile stress-strain response. To eliminate the deficiencies of sag effect, insufficient stiffness, and lack of durability for steel cable in long-span cable-stayed bridge, Wang et al.⁷³ proposed a carbon/basalt-HFRP cable, consisting of the core of basalt-FRP strands, viscoelastic material, inner sleeve, and the skin of carbon/basalt-HFRP strand. The test results exhibited that the carbon/basalt-HFRP cable could relieve the problems of large sag effect and imperfect mechanical properties relative to those of steel strand. Also, benefiting from the adjustable inherent damping of carbon/basalt-HFRP cable, it could present a better aerodynamic stability. Naito et al.^{74–76} tested the tensile properties of carbon/glass-HFRP bars with a pultruded core of carbon fiber and a braided sleeve of made glass fiber, wherein the volume fractions of carbon fiber ranged from 24.6% to 46.2%. It was observed that the stress acting on these carbon/glass-HFRP bars approximately linearly increased with strain up to their failure due to a higher carbon/glass ratio.

To present an ultimate strain higher than 3.0% and a strain hardening stage in the tensile stress-strain response of HFRP bar, Tepfers et al.⁷⁷ evaluated that the tensile properties of several types of HFRP bars, which were composed of a core of foam plastic and a thin cylindrical shell of glass fiber, a core of carbon fiber and a braided sleeve of aramid fiber, a core of carbon fiber and a braided sleeve of glass fiber, and a core of carbon fiber and a braided sleeve mingled aramid fiber with glass fiber, respectively. The experiment outcomes achieved the expected goal by using the braided fiber strands to

wrap a thin cylindrical fiber shell. Meanwhile, to ease the economic costs of inorganic fiber-reinforced polymer bars, Park et al.⁷⁸ utilized the natural-organic jute fiber instead of the different volume fractions of glass fiber to prepare glass/jute-HFRP bar. To make better bond strength between glass/jute-HFRP bar and concrete, a braided sleeve of PVA fiber was wrapped on the surface of glass/jute-HFRP bar. It could be concluded that the plastic deformation of glass/jute-HFRP bar occurred with an increasing substitution rate of jute fiber from 30% to 70%.

1.2 | Mingling of continuous fiber and steel rod/wire

As mentioned above, the tensile ductility of HFRP bar is realized by mingling two or more continuous fibers with different moduli of elasticity, which must comply with the principle of fiber-graded fracture. Hence, it would be inevitable that the high-modulus fiber has to exist in HFRP bar with a lower volume fraction. As a result, the tensile modulus of elasticity for ductile HFRP bar is lower than that of steel bar/wire under an affordable economic cost. Therefore, some researchers recommended mingling steel bar/wire with continuous fiber to produce steel/fiber-HFRP bar, which has a similar yield strain and modulus of elasticity to that of steel bar/wire. The steel bar/wire was typically located in the core of steel/fiber-HFRP bar, and it was protected by FRP shell from aggressive environment. Simultaneously, the FRP shell could also provide secondary stiffness for steel/fiber-HFRP bar after the core of steel bar/wire yielded. The steel bar/wire usually includes deformed steel bar, plain bar, steel wire, and steel strand; the continuous fiber is normally classified as aramid fiber, basalt fiber, carbon fiber, glass fiber, and natural fiber.

Wu et al.⁷⁹ tested the tensile properties of two types of steel/fiber-HFRP bars, and the inner steel bar was wrapped by the continuous carbon fiber or basalt fiber impregnated with polymer resin. They concluded that an ideal failure mode for these samples was obtained. First, the core of steel bar yielded within the gauge length of steel/carbon- and steel/basalt-HFRP bars. Subsequently, the wrapped sleeve of carbon fiber or basalt fiber was straightened and broke near the shrinkage section of steel bar. Finally, the yielded steel bar was pulled apart near the broken carbon fiber or basalt fiber. Consequently, the tensile stress-strain responses of steel/carbon- and steel/basalt-HFRP bars presented a stable secondary stiffness, an obvious yield point, and a considerable ultimate strain. Jung et al.⁸⁰ performed the tensile tests of steel/glass-HFRP bar, obtaining a bilinear tensile stress-strain

response with a strain hardening stage. Zhao et al.⁸¹ and Zhou et al.⁸² introduced two newly developed steel-FRP composite bars (SFCB) with an inner core of steel rod and a wrapper of CFRP or GFRP shell to construct different surface configurations. The results indicated that an inflection point presented in the tensile stress-strain response when the yield strain of inner steel rod was reached. Subsequently, the increasing load was mainly undergone by the outer FRP shell, which corresponded to a strain hardening stage of SFCB bar under tensile load. Sun et al.⁸³ identified that the complementary advantages for the constituent materials and construction morphology of steel/fiber-HFRP bar, composed of a core of steel bar and a wrapper of FRP shell. The outer FRP shell could provide a good protective effect for the inner steel bar from environment corrosion. Also, the outer FRP shell played an important role in ensuring a stable post-yielded tensile stiffness for steel/fiber-HFRP bar. Nanni et al.⁸⁴ took a steel/fiber-HFRP bar as the research object, which consisted of a core of steel bar and a braided shell of aramid fiber or vinylon fiber. To overcome the defects of brittle fracture and low modulus of elasticity for single-type FRP bar, Ma et al.⁸⁵ proposed a steel/fiber-HFRP bar with a core of steel bar and a braided sleeve of basalt fiber. Saikia et al.⁸⁶ prepared a new type of steel/fiber-HFRP bar, which was described that 10 roots of GFRP strands with a 2 mm diameter were wound helically on a 6 mm diameter steel rod. The experiment results confirmed three valuable information, including: a similar yield strain to that of steel rod, an ultimate strain equal to that of glass fiber, and a 102 GPa tensile modulus of elasticity. Seo et al.^{87,88} separately mingled steel rebar, plain bar, and steel wire with glass fiber to solve the brittle failure and a low modulus of elasticity for GFRP bar under tensile load.

Cui et al.^{89,90} manufactured a novel HFRP bar, consisting of a core of randomly dispersed steel fiber and glass fiber, and an interlayer mingling shell of Twaron fiber and carbon fiber. The carbon fiber was located on the outer layer of HFRP bar to protect the inner glass fiber from environmental erosion. The test results demonstrated the tensile ductility of this novel HFRP bar. Wang et al.⁵³ performed the tensile tests of steel/basalt-HFRP bars, containing the volume fractions of dispersed steel wire ranging from 20% to 40%. An approximate 1.0% yield strain was presented in the bilinear tensile stress-strain response, followed by a strain hardening stage corresponding to a large plastic deformation. In order to solve the brittleness fracture and reduce the economic cost for GFRP bar, the steel wires with different diameters were mingled with glass fiber at different volume fractions to prepare steel/glass-HFRP bars by Hwang et al.⁹¹ The results found that the tensile stress-strain

responses of steel/glass-HFRP bars nonlinearly developed with an increasing loading, and the tensile moduli of elasticity of steel/glass-HFRP bars were improved by 20% ~ 190% relative to that of GFRP bar.

Zhang et al.⁹² used the steel/glass-HFRP bar to improve the low modulus of elasticity and brittle failure of GFRP bar. A three-stage tensile stress-strain response was predicted for the steel/glass-HFRP bar as follows: a linear elastic stage corresponding to cooperative work between steel strand and glass fiber, a nonlinear stage corresponding to cooperative work between yielded steel strand and glass fiber, and a nonlinear stage corresponding to working alone for yielded steel strand after glass fiber failed. Hao et al.^{93,94} reported the tensile results of steel/glass-HFRP bar, indicating that the steel/glass-HFRP bar did not display yield characteristics similar to those of steel bar under tensile load. Because the internal steel strands were pulled off to cause the premature failure of steel/glass-HFRP bar. Fang et al.⁹⁵ performed a similar test for steel/glass-HFRP bar to obtain the same conclusion.

1.3 | Mingling of continuous fiber and the chopped-fiber randomly dispersed into polymer resin

To obtain the tensile ductility and reduce the economic cost of FRP bar, Behnam et al.^{57,58} designed and tested two novel types of HFRP bars, consisting of discontinuous fiber layer and continuous fiber layer. Based on the characteristics of constituent materials from outer layer to inner layer, the novel HFRP bars could be separated into two categories, including: (a) interlayer mingling of randomly-dispersed chopped-IM carbon fiber layer, randomly-dispersed chopped-LM carbon fiber layer, continuous aramid fiber layer, and continuous glass fiber layer (configuration A), and (b) interlayer mingling of randomly-dispersed chopped-IM carbon fiber layer, randomly-dispersed chopped-LM carbon fiber layer, continuous aramid fiber layer, and steel rod (configuration B), as shown Figure 6A,B. The results indicated that the purpose for solving the plastic failure of FRP bar was reached by mingling discontinuous fiber layer with continuous fiber layer. The failure sequence from outer layer to inner layer was observed for the HFRP bar with configuration A, resulting from an increasing fiber elongation from outer layer to inner layer regardless of discontinuous or continuous layer fiber. However, the inner core of steel rod first yielded, followed by the failure sequence from outer layer to inner layer for the HFRP bar with configuration B. A higher volume fraction of discontinuous fiber was not realized due to the difficulty of

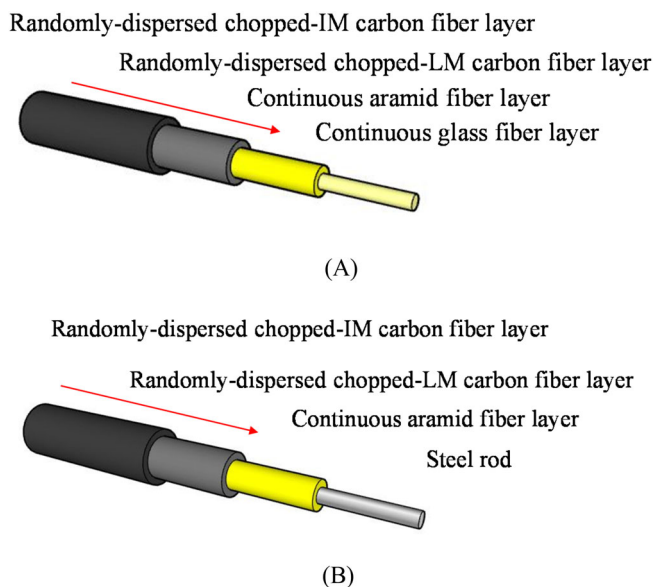


FIGURE 6 Mingling of continuous fiber layer and the randomly-dispersed chopped-fiber layer. (A) Configuration A. (B) Configuration B.

uniform dispersion for chopped fiber into polymer resin. Also, except for the volume fraction of chopped fiber, the tensile strength and modulus of elasticity for discontinuous carbon fiber layer were mainly dependent on the distribution orientation and uniformity of chopped carbon fiber in polymer resin.^{96,97} Consequently, although the addition of discontinuous fiber layers cut down the economic costs of HFRP bars, the obtained tensile strength and modulus of elasticity were dramatically reduced for the HFRP bars with configurations A and B. Meanwhile, it needed to be noted that a sharp load drop would occur at every failure of mingling fibers, which should be avoided during the design or application of HFRP bar-reinforced structural concrete.

1.4 | Mingling of aligned discontinuous fiber and continuous fiber

Ewen et al.⁹⁸ and Fahim et al.⁹⁹ designed two new types of FRP bars. The one was a single-type carbon-FRP bar, wherein the carbon fiber presented the coexistence states of discontinuous and continuous fibers. According to the distribution location of discontinuous carbon fiber, two configurations of single-type carbon-FRP bar were suggested, including: arrangement distribution around the circumference, and arrangement distribution along one side, as shown in Figure 7A,B. Due to ease of manufacture, the tensile properties of single-type carbon-FRP bar with the second configuration were tested. The results indicated that the failure mode of single-type carbon-FRP

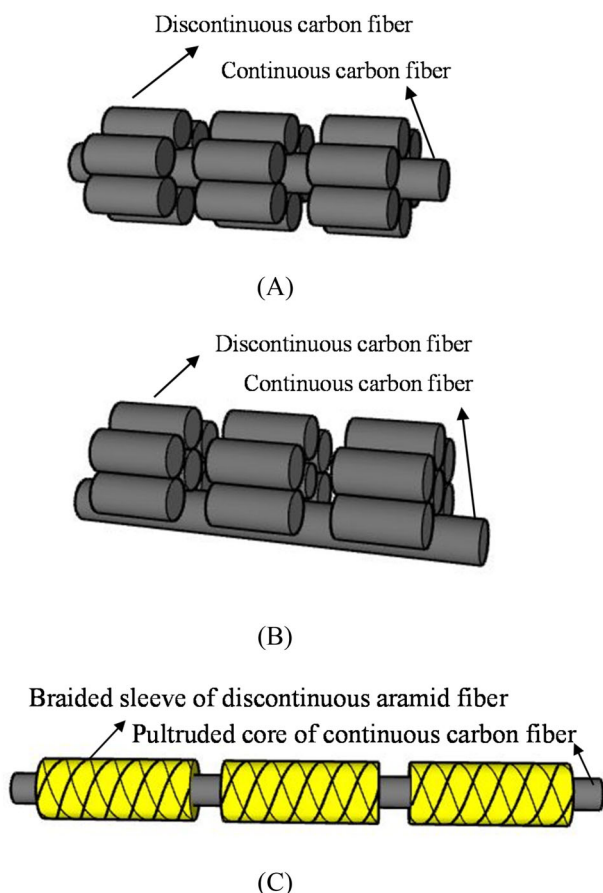


FIGURE 7 Scheme diagrams of HFRP bar mingled aligned discontinuous fiber with continuous fiber. (A) Arrangement distribution around the circumference. (B) Arrangement distribution along a side. (C) A discontinuous

bar with the second configuration showed the brittle fracture, though its structural characteristics had a promising potential.

The other was HFRP bar consisted of a braided sleeve of discontinuous aramid fiber and a pultruded core of continuous carbon fiber. The manufacturing procedure of this HFRP bar could be divided into two steps. First, the aramid/carbon-HFRP bar with a pultruded core of continuous carbon fiber and a braided sleeve of continuous aramid fiber was prepared by pultrusion and braiding processes. Second, the braided sleeve of continuous aramid fiber was cut circumferentially at an equal longitudinal length by a hand-held grinding wheel. The schematic diagram was presented in Figure 7C. Finally, the resulting HFRP bar was tested under tensile load, finding that the load-displacement response exhibited a multi-stage development trend belonging to plastic failure, as shown in Figure 8. Initially, the tensile stress went up linearly with the increase in strain in the elastic stage. Subsequently, when the friction shear strength (τ_f) between the braided sleeve of discontinuous aramid fiber and the core

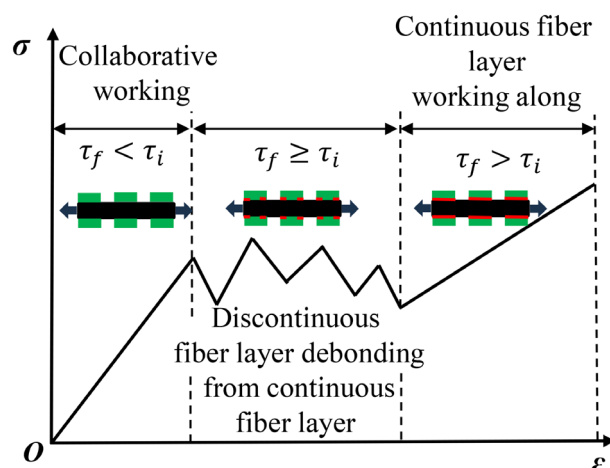


FIGURE 8 Tensile stress-strain response of discontinuous-continuous FRP bar.

of continuous carbon fiber approached to the lower limit of the interlaminar shear strength (τ_i) of HFRP bar, the braided sleeve of discontinuous aramid fiber began to debond from the pultruded core of continuous carbon fiber, which was accompanied by a dramatic stress drop. Next, when the friction shear strength between the braided sleeve of discontinuous aramid fiber and the core of continuous carbon fiber was larger than the lower limit of the interlaminar shear strength of HFRP bar, the remaining braided sleeves of discontinuous aramid fiber completely debonded from the pultruded core of continuous carbon fiber, which corresponded to multiple sudden drops in stress in the tensile stress-strain response. Ultimately, the pultruded core of continuous carbon fiber broke to cause the complete failure of carbon/aramid HFRP bar with the coexistence states of discontinuous and continuous fibers.

It can be concluded that the longitudinal lengths of discontinuous carbon fiber or braided sleeve of discontinuous aramid fiber were critical, which decided that the fracture or debonding of discontinuous fiber layer from the pultruded core of continuous fiber layer. The ideal design principle is that controlling the longitudinal length of discontinuous fiber layer makes it debond from the pultruded core of continuous fiber layer to obtain a notable tensile ductility. Meanwhile, it is worth being noted that the layer of discontinuous carbon fiber or the braided sleeves of discontinuous aramid fiber located on the outer of continuous/discontinuous carbon/aramid-HFRP bar is detrimental to their bond with concrete.

1.5 | Comprehensive evaluation

From the above review, it can be seen that the essential condition to realize the tensile ductility of HFRP bars

with two or more continuous fibers is to calculate the critical volume ratio of hybrid fibers based on the graded fracture theory of composite. However, the resulting HFRP bars with tensile ductility still have some issues, such as a large yield deformation, a low modulus of elasticity, and a sharp post-yielded stress sudden drop, as shown in Figure 9. These issues, coming from the failure procedure of ductile HFRP bars, are unfavorable to ductile HFRP bar-reinforced concrete components in service. Because the energy consumption mode of HFRP bar-reinforced concrete component is different from that of steel bar-reinforced concrete component under the application of external load, the energy consumption by the yield of steel bar accounts for 70% ~ 80% of the total energy for steel bar-reinforced concrete component, while this percentage declines to 10% ~ 20% for HFRP bar-reinforced concrete component. Meanwhile, a large deflection is inevitable for HFRP bar-reinforced concrete component under the application of external load, together with the storage of a large amount of strain energy. When the hybrid fibers happen graded fracture, the strain energy released can destroy HFRP bar-reinforced concrete component. Referring to the design method of FRP bar-reinforced concrete component at present, the method to solve a large deflection of HFRP bar-reinforced concrete component is to use excessive HFRP bars to reinforce concrete component. The excessive reinforcement ratio makes the failure of HFRP bar-reinforced concrete component initiate with the concrete crushing of compressive zone. This design principle is bound to contradict with the design philosophy for the structural concrete.

For the HFRP bar composed of steel bar/wire and continuous fiber, the stable secondary stiffness of steel/fiber-HFRP bars mainly depended on the characteristics of outer continuous fiber after the inner steel bar/wire

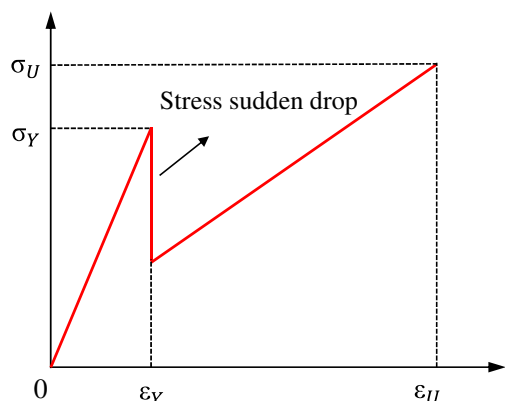


FIGURE 9 Typical tensile stress-strain response of ductile HFRP bar. (σ_Y , σ_U , ϵ_Y , and ϵ_U represent the yield stress, the ultimate stress, the yield strain, and the ultimate strain, respectively.)

yielded. On the one hand, the FRP sleeve wrapped around steel bar/wire could protect the steel bar/wire from environment corrosion, and it also provided the secondary stiffness for steel/fiber-HFRP bar after the inner steel bar/wire yielded. On the other hand, the inner steel bar/wire decided the yield strain and modulus of elasticity for steel/fiber-HFRP bar to make them similar to those of steel bar/wire. The yield of steel/fiber-HFRP bar would not cause the fiber delamination and post-yielded sudden drop in stress. However, it should be noted that the outer FRP sleeve cannot completely isolate the corrosion of inner steel bar/wire by external environment, while the steel/fiber-HFRP bar cannot satisfy the demands of non-magnetized or lightweight structural design.

For the HFRP bar mingled randomly-dispersed chopped-fiber with continuous fiber, this HFRP bar cannot be massively produced, and it cannot also replace steel bar to reinforce structural concrete in terms of the current production process and mechanical properties. Nevertheless, because of different damage mechanisms under tensile load, the HFRP bar consisted of aligned discontinuous fiber and continuous fiber could get rid of the deficiencies of HFRP bars composed of two or more continuous fibers. It can be regarded as a novel method to break the inherent shackles of ductile HFRP bars composed of two or more continuous fibers.

In this work, the design methods of tensile ductility for HFRP bars were systematically reviewed, and the shortcoming of these design methods were also pointed out. Subsequently, the graded fracture theory and preparation processes of HFRP bars were introduced. The solution to the deficiencies of ductile HFRP bars composed of two or more continuous fibers, including large yield deformation, low modulus of elasticity, and post-yielded sudden drop in stress, will be put forward.

2 | GRADED FRACTURE THEORY

As stated above, the volume ratio of hybrid fibers in the HFRP bar consisting of two or more continuous fibers must be controlled below a critical value, only in this way can the hybrid fibers with different elongations break sequentially under tensile load. This critical volume ratio of hybrid fibers in the ductile HFRP bar can be determined according to the graded fracture theory of composite as follows.

2.1 | Two types of hybrid fibers

For the ductile HFRP bar with two types of hybrid fibers,^{36,38,47} the critical value of volume ratio ($R_{L/H-critical}$) can be obtained as follows.

$$R_{L/H-critical} = \frac{V_{L,max}}{V_{H,min}} = \frac{E_H(\varepsilon_H - \varepsilon_L)}{E_L \varepsilon_L} \quad (1)$$

where $R_{L/H-critical}$ denotes the critical volume ratio of low elongation fiber to high elongation fiber; E , V , and ε are representative of tensile modulus of elasticity, volume fraction, and tensile strain of fiber, respectively; the subscripts L and H stand for the fiber with a low elongation and the fiber with a high elongation, respectively; max and min represent the maximum and minimum values.

Figure 10 displays the ideal stress-strain responses of ductile HFRP bars with two types of hybrid fibers under tensile load. When the volume ratio of low elongation fiber to high elongation fiber in HFRP bar is lower than the critical volume ratio determined by Equation (1), the tensile stress-strain response of HFRP bar presents the development trend of curve one. The curve one consists of three stages, including a linear elastic stage, a platform stage, and a strain hardening stage. Under tensile load, the two types of hybrid fibers in HFRP bar are initially stressed together until the low elongation fiber first reaches its ultimate strain and begins to break, that is, the HFRP bar yields at the point one. Because of the bridge effect of neighboring fiber with a high elongation, the interval distribution of tensile strength for fiber with a low elongation, and the shear lag effect of composite,⁹⁹ then the damage of low elongation fiber gradually accumulates with the increase in load up to the point two, and a corresponding platform stage appears on the tensile stress-strain response. Subsequently, only high elongation fiber in the HFRP bar undergoes an increasing load alone until its ultimate strain is reached, and the HFRP bar is destroyed at the point three, which corresponds to a strain hardening stage in the tensile stress-strain response of curve one.

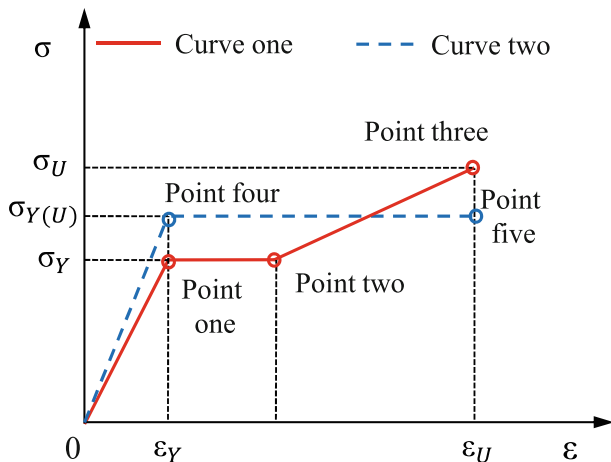


FIGURE 10 Ideal stress-strain curves of HFRP bar mingled by two types of hybrid fibers.

When the volume ratio of low elongation fiber to high elongation fiber in HFRP bar is equal to the critical volume ratio determined by Equation (1), the tensile stress-strain response of HFRP bar shows the development trend of curve two. The curve two is composed of two stages, including a linear elastic stage and a platform stage. Comparing to the characteristics of curve one, the reason for the absence of a strain hardening stage in the curve two is that the high elongation fiber is just capable of undergoing the load released by the broken low elongation fiber, and unable to undergo a further increasing load after the low elongation fiber broke.

2.2 | Three types of hybrid fibers

For the ductile HFRP bar with three types of hybrid fibers,⁹⁰ the critical values of volume ratios ($R_{L/M-critical}$ and $R_{M/H-critical}$) can be obtained by Equations (2) and (3).

$$R_{L/M-critical} = \frac{V_{L,max}}{V_{M,min}} = \frac{E_M \varepsilon_H (\varepsilon_M - \varepsilon_L)}{E_L \varepsilon_L (\varepsilon_H - \varepsilon_M)} \quad (2)$$

$$R_{M/H-critical} = \frac{V_{M,max}}{V_{H,min}} = \frac{E_H (\varepsilon_H - \varepsilon_M)}{E_M \varepsilon_M} \quad (3)$$

where $R_{L/M-critical}$ is the critical volume ratio of low elongation fiber to middle elongation fiber; $R_{M/H-critical}$ denotes the critical volume ratio of middle elongation fiber to high elongation fiber; the subscript M stands for the fiber with a middle elongation.

Then V_L , V_M , and V_H can be worked out by combining Equations (2)–(4) to ensure that the HFRP bar with three hybrid fibers will have a three-step fracture.

$$V_L + V_M + V_H = 1 \quad (4)$$

In addition, substituting the resulting V_L and V_H to Equation (5) can also verify whether the result is correct or not.

$$R_{L/H-critical} = \frac{V_{L,max}}{V_{H,min}} = \frac{E_H \varepsilon_H (\varepsilon_H - \varepsilon_L)}{E_L \varepsilon_L \varepsilon_M} \quad (5)$$

where $R_{L/H-critical}$ is representative for the critical volume ratio of low-elongation fiber to high elongation fiber.

Figure 11 presents two ideal stress-strain responses of HFRP bar with three types of hybrid fibers under tensile load. When the volume ratio of low elongation fiber to middle elongation fiber, the volume ratio of low elongation fiber to high elongation fiber, and the volume ratio of middle elongation fiber to high elongation fiber inside

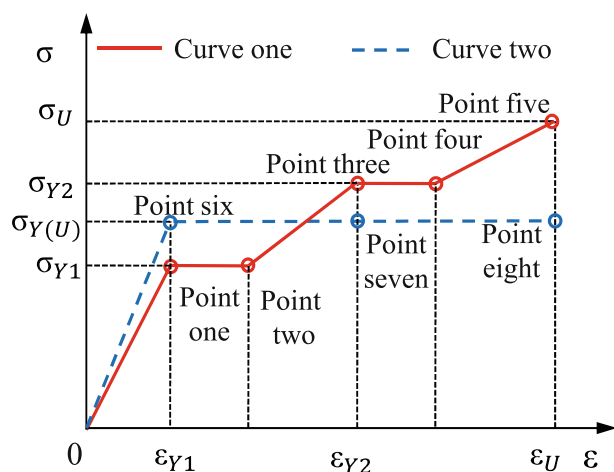


FIGURE 11 Ideal stress-strain curves of HFRP bar mingled by three types of three fibers.

HFRP bar are less than their critical volume ratios determined by Equations (2) to (5), the tensile stress-strain response of HFRP bar with three types of hybrid fibers shows the development trend of curve one in Figure 11. The curve one consists of five stages, a linear elastic stage, two platform stages, and two strain hardening stages. As shown in Figure 11, the point one represents that the low elongation fiber in HFRP bar begins to break, and the point two signifies that the low elongation fiber withdraws from work completely. The platform stage between the point one and point two is representative for the damage accumulation of low elongation fiber under tensile load, which comes from the bridge effect of middle and high elongation fibers, the interval distribution of tensile strength for low elongation fiber, and the shear lag effect of composite. The strain hardening stage between the point two and point three originates from both middle elongation fiber and high elongation fiber working together to undergo a further increasing load. Similarly, the point three is the starting point from which the middle elongation fiber begins to fracture, and the point four represents the end of work for the middle elongation fiber. The platform between the point three and point four is representative for the damage accumulation of middle elongation fiber under tensile load, which is from the bridge effect of high elongation fiber, the interval distribution of tensile strength for middle elongation fiber, and the shear lag effect of composite. Subsequently, the high elongation fiber undergoes alone the increasing load corresponding to the strain hardening stage between the point four and point five until the ultimate strain of high elongation fiber (point five) is reached. Finally, the HFRP bar is destroyed.

When the volume ratio of low elongation fiber to middle elongation fiber, the volume ratio of low

elongation fiber to high elongation fiber, and the volume ratio of middle elongation fiber to high elongation fiber in HFRP bar are equal to the critical volume ratios of the three hybrid fibers determined by Equations (2) to (5), the tensile stress-strain response of HFRP bar with three types of hybrid fibers presents the development trend of curve two in Figure 11. The curve two includes two stages, a linear elastic stage and a platform stage. Compared with the variation tendency of curve one in Figure 11, the reason for the loss of the two strain hardening stages in the curve two is that the middle elongation fiber and high elongation fiber are unable to undergo a further increasing load after they separately share the transferred load from the failed low elongation fiber.

3 | PRODUCTION PROCESS OF HFRP BAR

At present, the production process of HFRP bars mainly included pultrusion process and braiding process, while the winding process is used to form the serrated ribs on the surface of pultruded HFRP core.

3.1 | Pultrusion and winding processes

Figure 12 depicts the schematic diagrams of the newest pultrusion-winding processes.^{100,101} Compared with the previous pultrusion-winding processes, the newest pultrusion-winding processes have been improved in following aspects: (1) the addition of constant temperature water bath to maintain the fluidity of resin during the process of preparation; (2) the addition of front preheating box to increase the wettability of resin on the surface of fibers; (3) the addition of secondary extrusion device to further promote the uniform distribution of resin among fibers; (4) the change of box-type thermal treatment devices, including post-heating box, gel box, and curing box, and their working temperatures increase in a gradient to prevent the internal defects of FRP bars caused by sudden increase of temperature; (5) the addition of cooling box, which is a transitional device connecting high-temperature box and in-site environment, and it can protect the high-temperature FRP bars from in-site environment to cause shrinkage micro-cracks. It has been verified by experiments that the newest pultrusion-winding processes can significantly reduce the discreteness of products.^{102–105}

Figure 13 presents the schematic diagram of pultrusion device and winding device in the pultrusion-winding

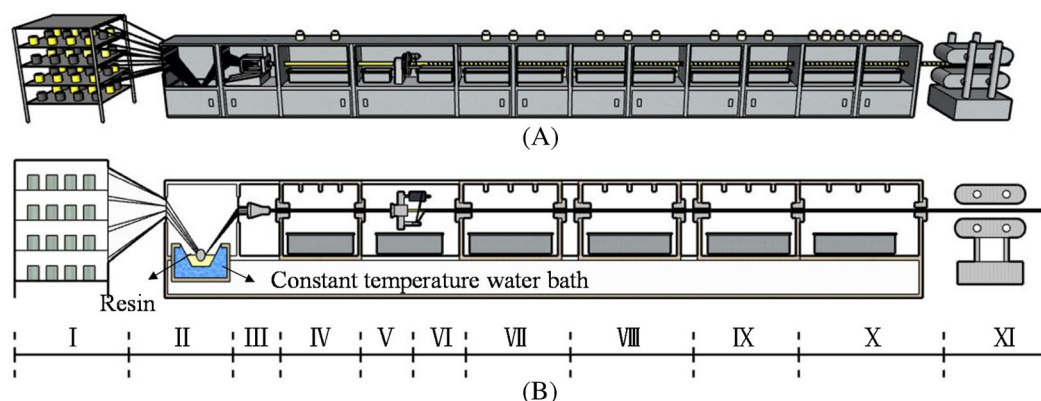
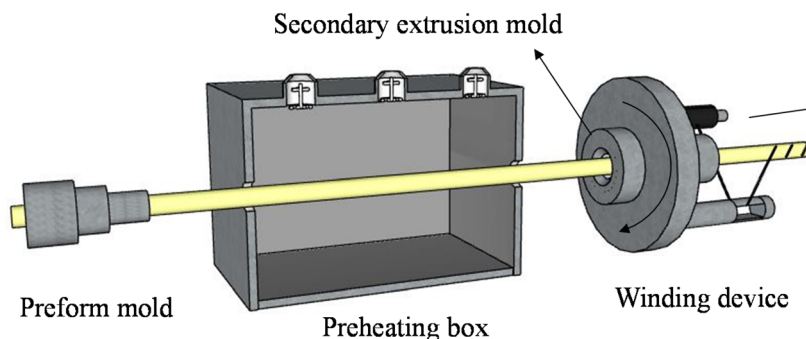


FIGURE 12 Schematic diagrams of pultrusion and winding processes. (A) Three-dimensional picture; (B) Section picture. I. Yarn frame; II. Impregnation device; III. Preform mold; IV. Front preheating box; V. Secondary extrusion mold; VI. Winding device; VII. Post preheating box; VIII. Gel box; IX. Curing box; X. Cooling box; XI. Tractor.

FIGURE 13 Pultrusion device and winding device.



processes of FRP bars.^{106,107} As shown in Figure 13, the fiber bundles impregnated with resin are first integrated into a whole by preforming mold. Then the resin matrix is evenly redistributed among fibers through the preheating box and the secondary extrusion mold. Subsequently, a row of spiral serrated ribs is formed on the surface of pultruded FRP bar by the winding device. The rib depth and rib pitch can be controlled by adjusting the tightness of winding tape and the speed of winding device. Finally, the preformed FRP bar is thermally treated and fixed-scale cutting to obtain the target product.

3.2 | Pultrusion and braiding processes

Figure 14 exhibits the schematic diagram of pultrusion-braiding processes for FRP bars.⁸⁷ Compared with the pultrusion-winding processes of FRP bar, the braiding process is introduced into the pultrusion process, and a braided fiber layer can fabricate on the surface of pultruded core. The braided layer is helpful to improve the bond behavior of FRP bar with concrete and increase the ultimate strain of FRP bar.

4 | FUTURE PROSPECTS

Currently, the design concept of HFRP bar is from the perspective of tensile ductility. Nevertheless, the design method based on the graded fracture theory of composite will inevitably result in a lower tensile modulus of elasticity for HFRP bar. Furthermore, the ultimate strain of economic fibers and the failure mode of monofilament fiber have to bring about a large yield strain and post-yielded stress sudden drop in the stress-strain response of HFRP bar under tensile load. Consequently, a new-type fiber with a lower elongation approaching to the yield strain of steel bar is urgently needed. However, in the early stage of emergence of new-type fiber, its cost is probably unaffordable and the application of new-type HFRP bar into civil engineering is unpredictable. As a result, it could be a good solution to change the failure mode of FRP bar under tensile load from the production technology to obtain the mechanical properties similar to those of steel bar.

The types of fiber composite are diverse based on the kind of matrix, including metal, cement, polymer, ceramic, and so forth. Herein the fiber-reinforced polymer bars are mainly introduced. Although the superior

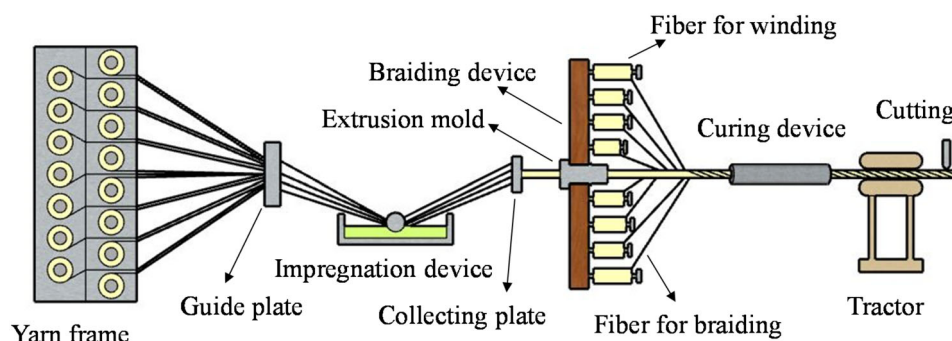


FIGURE 14 Pultrusion-braiding processes of FRP bar.

mechanical properties of FRP composites are typically achieved by continuous fiber-reinforced polymer matrix, the catastrophic failure of continuous fibers will bring about great disadvantages for the structural design and functional application in civil engineering.^{108,109} As a solution, the high mechanical properties of highly aligned discontinuous fiber-reinforced composites make it an ideal candidate.^{110,111} Furthermore, the deformation and slip of highly aligned discontinuous fiber layer may be create a ductile response for single-type FRP bar under tensile load.^{112–114}

In recent decades, the highly aligned discontinuous fiber-reinforced polymer composites have been studied by many researchers.^{115–118} Pickering et al.¹¹⁹ pointed out that the tensile modulus of elasticity and strength for aligned discontinuous natural fiber composites were significantly higher than those of randomly-dispersed chopped-natural fiber composites. Bolk et al.¹²⁰ declared that the superior performance of aligned discontinuous fiber-reinforced composites could compete with that of continuous fiber-reinforced composites. Yu et al.¹²¹ successfully produced aligned discontinuous carbon fiber-reinforced epoxy composites, of which the length of the aligned discontinuous carbon fiber was 3 mm. The outcomes indicated that when the total volume fraction of discontinuous carbon fiber was 55%, the discontinuous carbon-FRP bar possessed a 115 GPa tensile modulus of elasticity and a 1509 MPa tensile strength. To exhibit a gradual failure mode instead of a catastrophic drop in stress for intermingled carbon/self-reinforced polypropylene hybrid composite under loading, Tang et al.¹²² introduced and tested the mechanical properties of hybrid composites consisting of highly aligned discontinuous carbon fiber layer and continuous polypropylene layer. They concluded that the effect of highly aligned discontinuous carbon fiber layer on the initial modulus of elasticity was marginal, and it could help to reduce post-yielded sudden drop in stress. Meanwhile, the stress concentration at the end of discontinuous carbon fiber caused the debonding of discontinuous carbon fiber layer at a lower failure strain, which corresponded to the yield

strain of this discontinuous hybrid composite. Additionally, Bowyer et al.¹²³ also considered that the shorter discontinuous fiber could not be strained synergistically with polymer matrix, which would result in the failure of discontinuous fiber composite in a lower strain to prevent its potential mechanical properties. Henry et al.¹²⁴ conducted on a semi-analytical simulation for orientated discontinuous composites, observing that the failure modes of orientated discontinuous composites were dependent on the overlap length of short fiber and the strength of long fiber in composites.

Additionally, some researchers also tested the hybrid composite of highly aligned discontinuous fiber layer and continuous fiber layer.¹²⁵ Czel et al.¹²⁶ demonstrated the feasibility of discontinuous carbon fiber and continuous glass fiber interlayer hybrid reinforced composites. They pointed out that a nonlinear tensile stress-strain response could be obtained, which consisted of a linear elastic stage, a platform stage, and a strain hardening stage. Moreover, Yu et al.^{127,128} also obtained a similar result with the hybrid composites composed of highly aligned discontinuous carbon fiber and glass fiber. In addition, Yu et al.¹²⁹ also proposed a novel hybrid composite of the highly aligned discontinuous carbon fiber layer located between continuous S-glass fiber layers to balance tensile modulus, strength, and ductility of hybrid composite. Tang et al.^{130,131} also gave the same viewpoint and stated that the results were derivate from the change of failure mode for hybrid composite composed of discontinuous carbon fiber layer and self-reinforced polypropylene layer.

Therefore, the authors are inspired from the design in the hybrid composite of highly aligned discontinuous fiber layer and continuous fiber layer, and promote the application of this similar configuration to FRP bar, as shown in Figure 15. This idea is mainly based on two current practical problems: (1) Characteristics of low modulus of elasticity, high yield strain, and post-yielded sudden drop in stress hinder the popularization of continuous-type HFRP bars with tensile ductility in a large number of projects; (2) because of the production

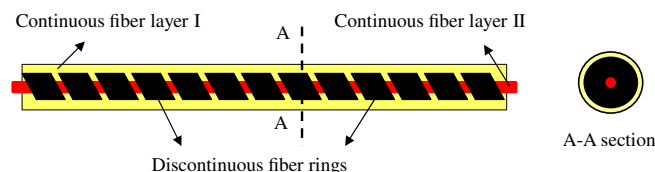


FIGURE 15 Schematic diagram of discontinuous-continuous fiber-reinforced polymer bar.

cost, the future development trend of carbon fiber is toward large fiber bundle. For example, a bundle of carbon fiber containing the number of monofilament fiber will increase from the current 1, 3, 6, 12 k and 24 to 50 k, 60, 120, 360 k, and so forth. It needs to consider that whether the existence of large carbon fiber bundle will bring the obstacles to the design and production of HFRP bars based on the graded fracture theory of composite in the future, owing to a large bundle of carbon fiber corresponding to a large cross-sectional area. In summary, the authors suggest solving the current problems of HFRP bar from its failure mode under tensile load.

5 | CONCLUSIONS

In this paper, the ductility design methods for FRP bar were reviewed, which mainly included mingling of two or more continuous fibers, mingling of continuous fiber with steel bar/wire, mingling of continuous fiber with chopped-fiber randomly dispersed into polymer resin, and mingling of highly aligned discontinuous fiber with continuous fiber. It was found that except for a limited ultimate strain, the main obstacles are low tensile modulus of elasticity, high yield strain, and post-yielded sudden drop in stress for the ductile HFRP bars based on the graded fracture theory, which hinder their large-scale application in civil engineering. Currently, these problems cannot be overcome within the range of affordable economic cost through changing the type of continuous fiber.

A new design concept of based on single-type FRP bar was proposed by the authors to solve the current defects of ductile HFRP bar based on the graded fracture theory. In this novel single-type FRP bar, the highly aligned discontinuous fiber and the same continuous fiber coexist in single-type FRP bar. The failure mode of this novel FRP bar is different from that of the ductile HFRP bar designed based on the graded fracture theory. The tensile ductility of discontinuous/continuous single-type FRP bar originates from the debonding and stable pull-out of discontinuous fiber layer under an increasing load. As a result, the post-yielded sudden drop in stress

can be removed for the ductile HFRP bar designed based on the graded fracture theory. Also, the yield strain can be controlled by adjusting the length of discontinuous fiber layer. Meanwhile, the authors recommend the carbon fiber being used as the reinforced phase of this discontinuous/continuous single-type FRP bar with tensile ductility, on account of its high tensile modulus of elasticity and a relatively affordable economic cost.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Yu Zhang  <https://orcid.org/0000-0001-7115-1082>

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